BEAM ANALYSIS by the STIFFNESS METHOD

Only Pin Supports and Point Loads are included here. Distributed Loads are not included yet.
Does not include provisions for applied moments yet.

Matrix programs / routines based on ORIGIN =1.

version: beta v.2026.07.20

PROBLEM DESCRIPTION

Problem description & details summary can be input in this box.

Problem Description (short): $problem := \text{"single span, 3 point loads"}$

Mat

Begin beam positions:
(always ZERO) $x_0 := 0\ ft$

MM

End beam position (beam length): $x_{end} := 50\ ft$

Total Beam Length $L_{beam} := x_{end} = 50\ ft$

Number (qty) of result points
printed along elements: $n_r := 10$

INPUT - MATERIAL

Material Type: $material :=$ steel

Modulus of Elasticity: $E := material_2 = \left(2.9 \cdot 10^4\right)\ ksi$

Section Description: $section := \text{"W14x61"}$

Moment of Inertia of Beam: $I := 640\ in^4$

INPUT - SUPPORT LOCATION
All supports assumed to be pins at this time.

Support Position Table

x_{supp} (ft)
0
25
50

Support Positions: $x_{supp}{}^{T} = [0\ \ 25\ \ 50]\ ft$

Number of Supports: $n_{supp} := \mathrm{rows}(x_{supp}) = 3$

INPUT - POINT LOAD TABLE
Enter any number of point loads. Negative forced acting downward.

Working Values
Truck Axle Load(s) - UNITLESS: $P_a := 8$ $x_1 := 7.5$ $x_2 := x_1 + 6$

Point Load Input Table:

(can insert rows below to add more loads)

x_{Pload} (ft)	P_x (lbf)	M_x ($lbf \cdot ft$)
12.5	−5000	0

$check\_point :=$ "check to make sure all loads are less than the beam length"

INPUT - DISTRIBUTED LOAD TABLE
Enter any number of distributed loads, maintaining the (R rows x columns)

DISTRIBUTED LOADS ARE NOT INCLUDED IN THE ANALYSIS YET!

Distributed Load Input Table:

(can insert rows below to add more loads)

x_{wstart} (ft)	x_{wend} (ft)	w_{start} $\left(\frac{kip}{ft}\right)$	w_{end} $\left(\frac{kip}{ft}\right)$
0	4	−1	−1

NOT INCLUDED IN ANALYSIS YET!!!

$check_dist := $ “check to make

WORKING EQUATIONS - NON-PRINTING AREA

REACTION OUTPUT EQUATIONS - NON-PRINTING AREA

SHEAR OUTPUT EQUATIONS - NON-PRINTING AREA

MOMENT OUTPUT EQUATIONS - NON-PRINTING AREA

OUTPUT - SUPPORT REACTIONS

$$Reactions_{node} := \begin{bmatrix} \text{“Node”} & \text{“position”} & \text{“vert. reaction”} \end{bmatrix}$$

Node, position, and vert. reaction: $Reactions_{node}{}^{T} = \begin{bmatrix} 1 & 3 & 4 \\ 0\ m & 7.62\ m & 15.24\ m \\ \left(9.035 \cdot 10^{3}\right)\ N & \left(1.529 \cdot 10^{4}\right)\ N & -2.085 \cdot 10^{3}\ N \end{bmatrix}$

$$ReacOut_{node}{}^{T} = \begin{bmatrix} 1 & 3 & 4 \end{bmatrix}$$

$$ReacOut_{pos}{}^{T} = \begin{bmatrix} 0 & 25 & 50 \end{bmatrix}\ ft$$

$$ReacOut_{force}{}^{T} = \begin{bmatrix} 2.031 & 3.438 & -0.469 \end{bmatrix}\ kip$$

OUTPUT - SHEAR FORCE

THIS ANALYSIS DOES NOT YET INCLUDE ANY DISTRIBUTED LOADS.

$$V_{node} := [\text{“Node”} \ \ \text{“position”} \ \ \text{“Left force”} \ \ \text{“Right force”}]$$

Shear forces at nodes:

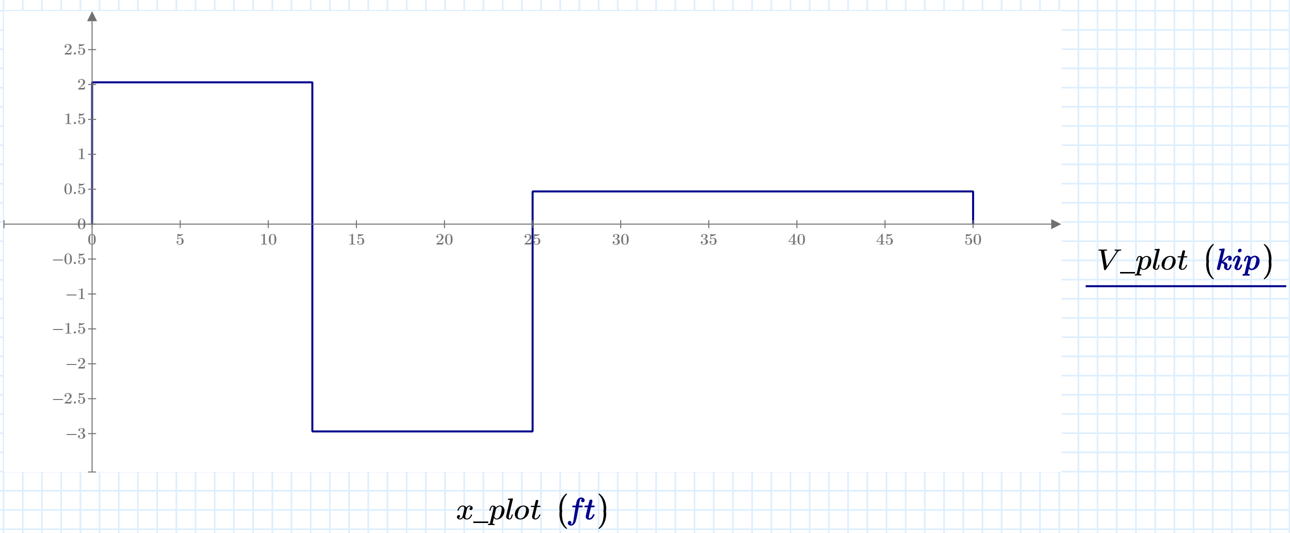
$$V_{node} = \begin{bmatrix} 1 & 0\ m & 0\ N & (9.035 \cdot 10^{3})\ N \\ 2 & 3.81\ m & (9.035 \cdot 10^{3})\ N & -1.321 \cdot 10^{4}\ N \\ 3 & 7.62\ m & -1.321 \cdot 10^{4}\ N & (2.085 \cdot 10^{3})\ N \\ 4 & 15.24\ m & (2.085 \cdot 10^{3})\ N & -3.183 \cdot 10^{-12}\ N \end{bmatrix}$$

$$Vout_{node}^{\mathrm{T}} = [1 \ \ 2 \ \ 3 \ \ 4]$$

$$Vout_{pos}^{\mathrm{T}} = [0 \ \ 12.5 \ \ 25 \ \ 50]\ ft$$

$$Vout_{left}^{\mathrm{T}} = [0 \ \ 2.031 \ \ -2.969 \ \ 0.469]\ kip$$

$$Vout_{right}^{\mathrm{T}} = [2.031 \ \ -2.969 \ \ 0.469 \ \ -7.156 \cdot 10^{-16}]\ kip$$



OUTPUT - MOMENT

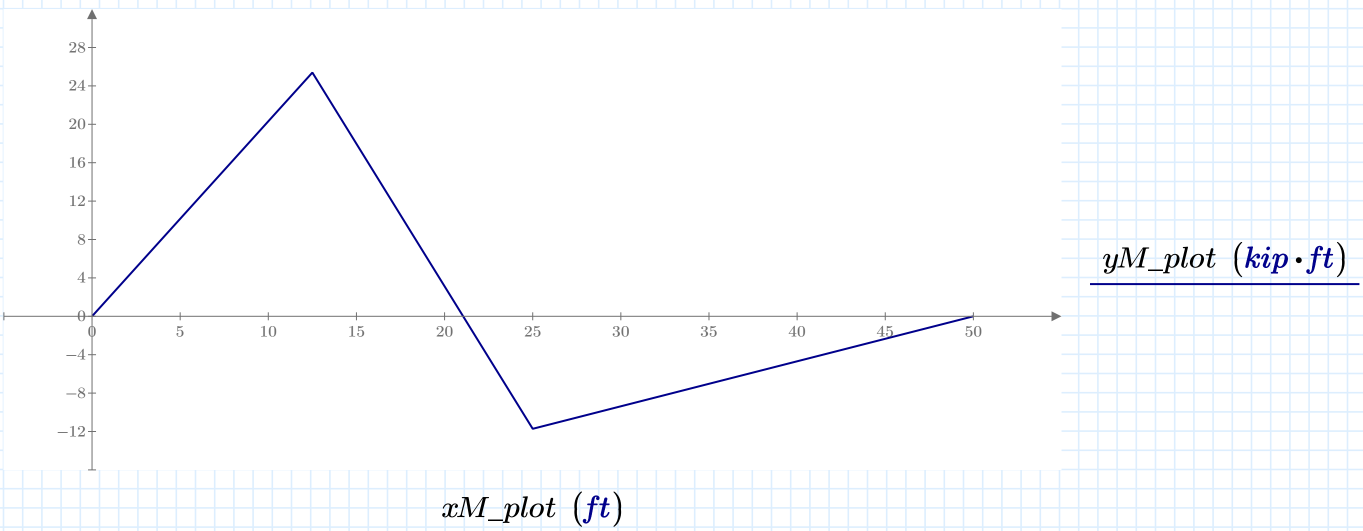
THIS ANALYSIS DOES NOT YET INCLUDE ANY DISTRIBUTED LOADS.

Specific / sample point: $x_s := 6.25\ ft$

Moment value at specified point: $Mpnt\left(x_s\right) = 12.695\ kip \cdot ft$

Maximum Moment: $M_{max} := \max\left(Mout\right) = 25.391\ kip \cdot ft$

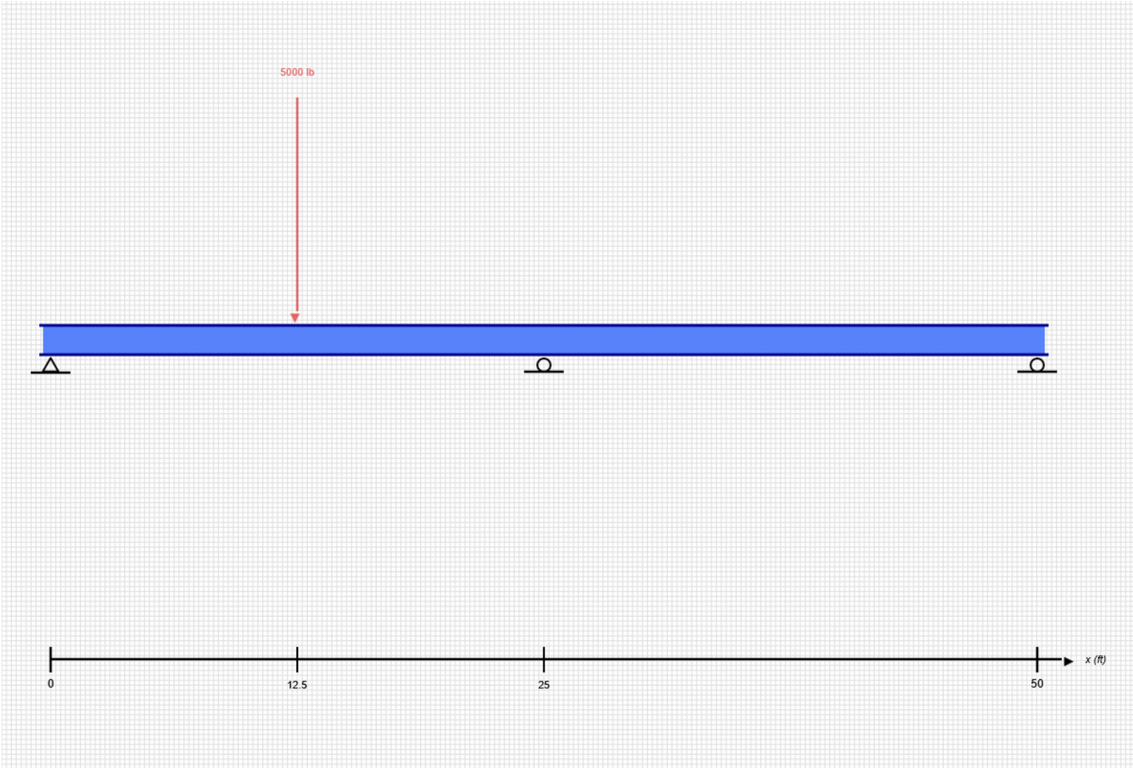
Minimum Moment: $M_{min} := min\left(Mout\right) = -11.719\ kip \cdot ft$



SkyCiv
CLOUD STRUCTURAL SOFTWARE

SKYCIV BEAM ANALYSIS REPORT

Load Combination: DL



Software: SkyCiv Beam v3.3.2
Thu Jun 18 2026 11:25:37 GMT-0400 (Eastern Daylight Time)

Included in this Report:

- Input Summary
- Beam Section
- Free Body Diagram (FBD)
- Analysis Summary
- Analysis Results
- Bending Moment Diagram (BMD)
- Shear Force Diagram (SFD)
- Deflection Results

INPUT SUMMARY

General Info

Beam Length:	50 ft
Section Name:	W14x61
Self Weight:	False

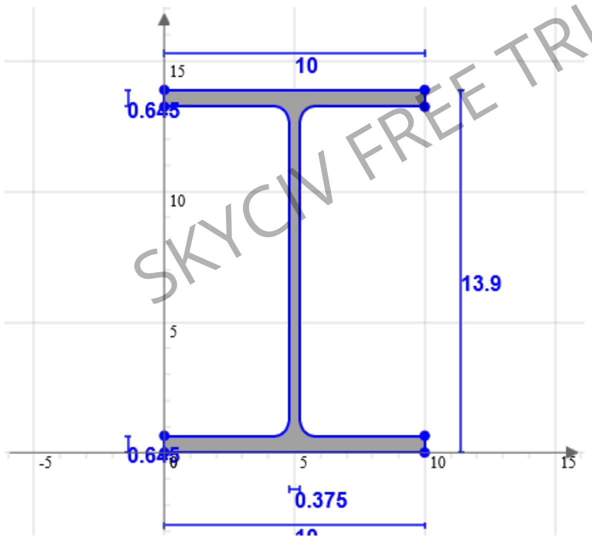
Supports

Support Type	Location
Pinned	0 ft
Roller	50 ft
Roller	25 ft

Loads

Load Type	Location	Magnitude	Load Case
Point Load	12.5 ft	-5000 lb	DL

Beam Section



Geometric Properties		
A	17.9	in^2
C_z	5	in
C_y	6.95	in

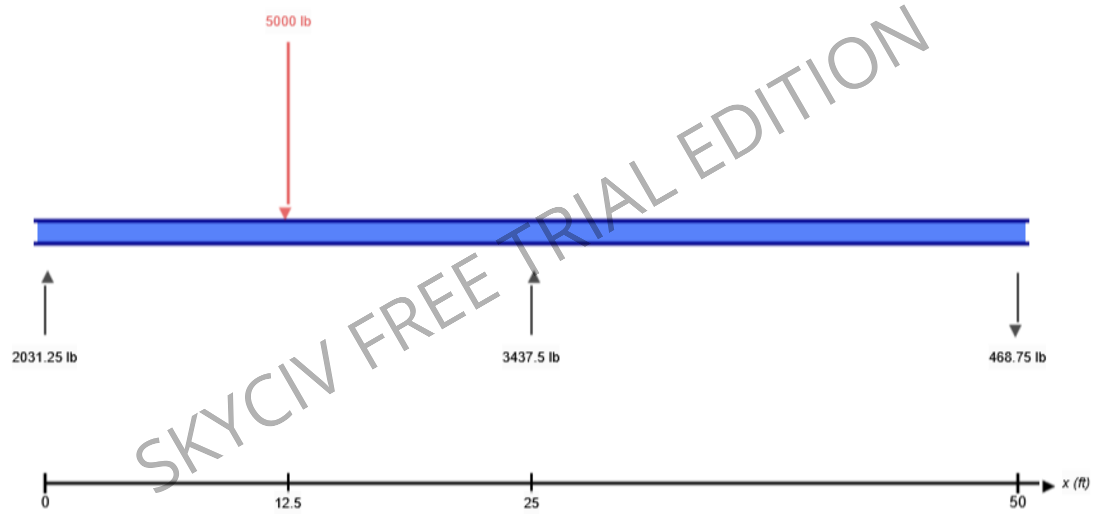
Bending Properties		
I_z	640	in^4
I_y	107	in^4

Shear Properties		
A_z	11.277	in^2
A_y	4.99	in^2

Torsion Properties		
J	2.19	in^4
r	0.991	in

Shape	Material	E (ksi)	v	ρ (lb/ft^3)
W14x61	Structural Steel	29000	0.27	490

FREE BODY DIAGRAM



RESULT SUMMARY

Check	Status	Limit	Ratio	Max
Deflection	PASS	L/250	0.045	L/5508
Custom Stress Limit	PASS	39 ksi	0.085	3.309 ksi
Material Yield	PASS	36 ksi	0.092	3.309 ksi
Material Strength	PASS	58 ksi	0.057	3.309 ksi

ANALYSIS RESULTS

Reactions

Support at	R_x (lb)	R_y (lb)	M (kip-ft)
0	0	2031.25	0
25	0	3437.5	0
50	0	-468.75	0

Force Extremes

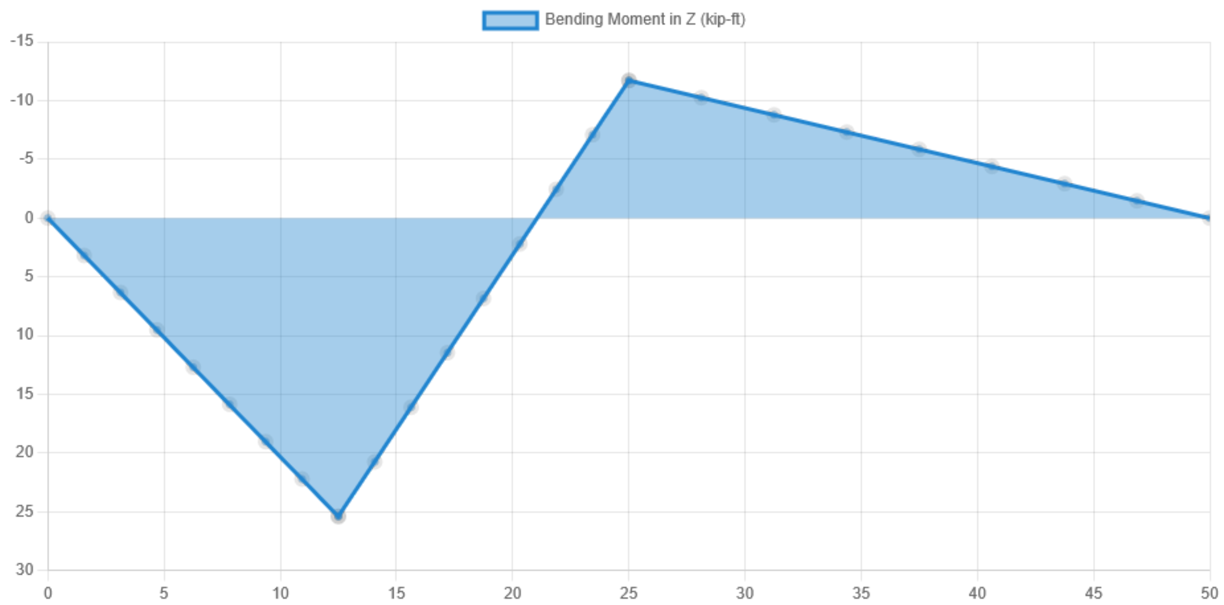
Result	Max	Min
Bending Moment	25.391 kip-ft	-11.719 kip-ft
Shear	2031.25 lb	-2968.75 lb
Displacement Y	0.043 in	-0.109 in

Stress Extremes

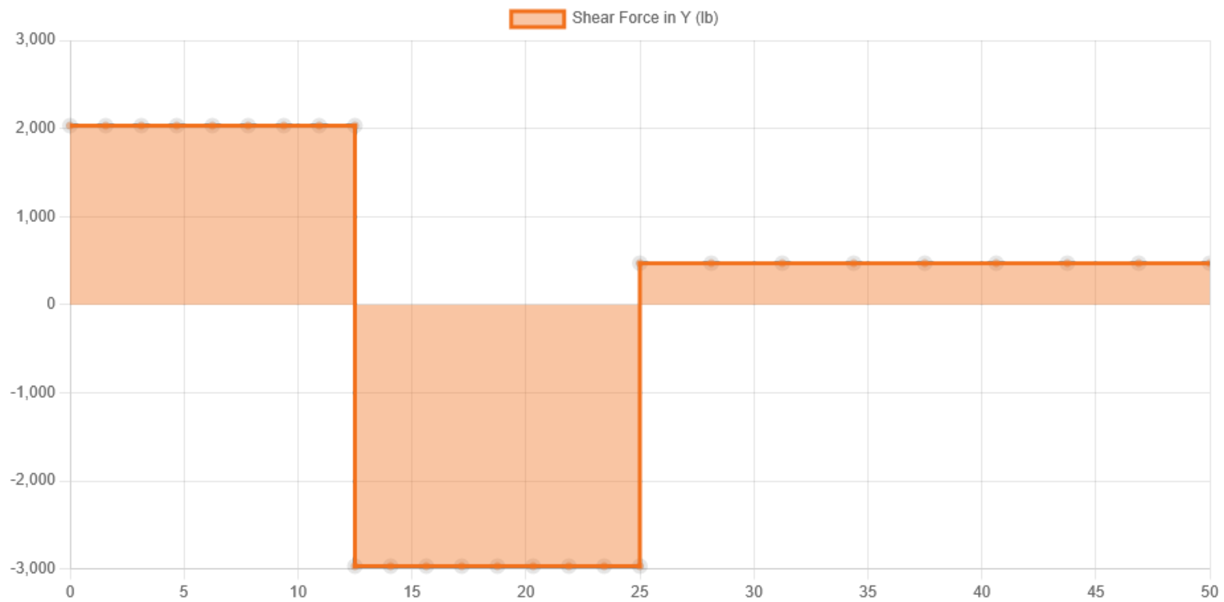
Result	Max (ksi)	Min (ksi)
Bending Stress	3.309	-3.309
Shear Stress Total	0.634	0.1
Max Combined Normal Stress	3.309	0
Min Combined Normal Stress	0	-3.309

DIAGRAMS

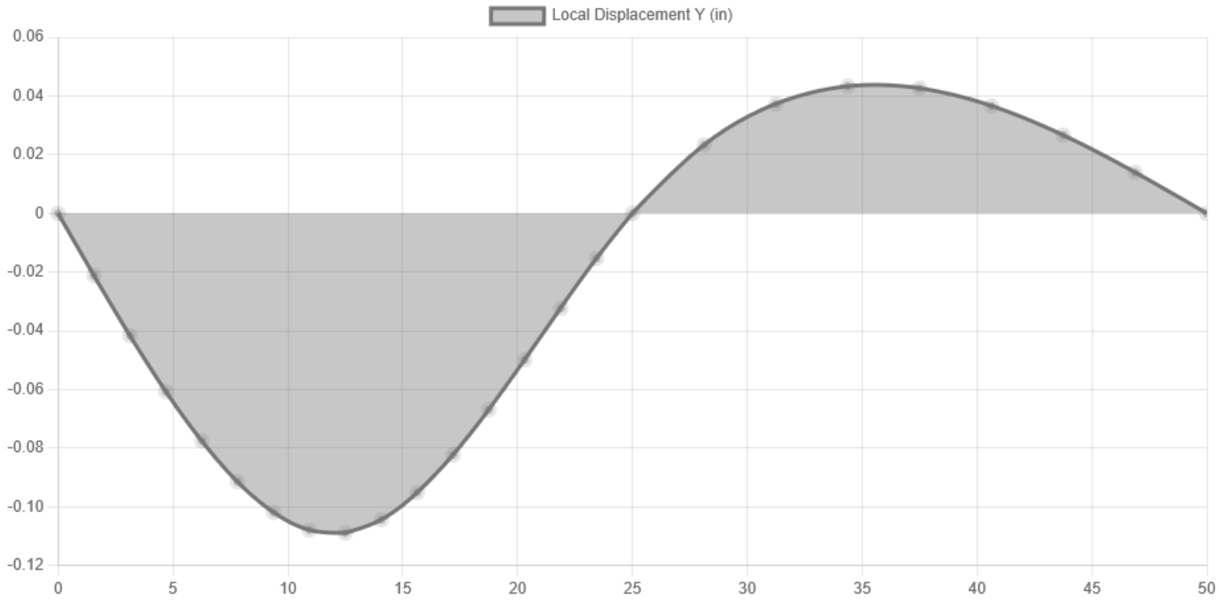
Bending Moment Diagram



Shear Force Diagram



Displacement



Location (ft)	0	12.5	25	34.375	50
Total Deflection (in)	0	0.109	0	0.043	0
Deflection/Span	-	L/2754	-	L/6930	-

The Deflection/Span results are calculated using the analysis results and the Deflection Limit of L/250 set in the **model settings.**